

Naam:

Studentnummer:

PNEUM-T1

Opgave 1 – Perslucht

a)

Druk, volume, temperatuur, dichtheid en/of luchtvochtigheid.

3/5 = 1P

b)

Lucht is samendrukbaar, olie (bijna) niet.

1P

c)

① $30^\circ\text{C} \approx 30 \text{ g/m}^3$

① $30 \text{ g/m}^3 \cdot 3 \text{ m}^3 \cdot 80\% = 72 \text{ g}$

① $72 \text{ g} - 30 \text{ g/m}^3 = 42 \text{ g}$ ① H_2O of Bodem

4P

d)

$$F = PA \quad P = \frac{F}{A}$$

① $\frac{50000 + 200}{\frac{1}{4}\pi \cdot 0,2^2} = \frac{50200}{0,0314} = 1.597 \text{ g/1P}$

① 15.9 BAR ①
of
 1598 kPa

3P

e)

$$F = P \cdot A \quad A = \frac{F}{P} \quad F_c = \frac{F}{2} = \frac{1700}{2} \textcircled{1}$$

$$A = \frac{\frac{1700}{2}}{6.0 \cdot 10^5} = 0.00141667$$

$$A = \frac{1}{4} \pi D^2 \quad D = \sqrt{\frac{4A}{\pi}} \Rightarrow D = 42 \text{mm} \textcircled{1} \quad \underline{3P}$$

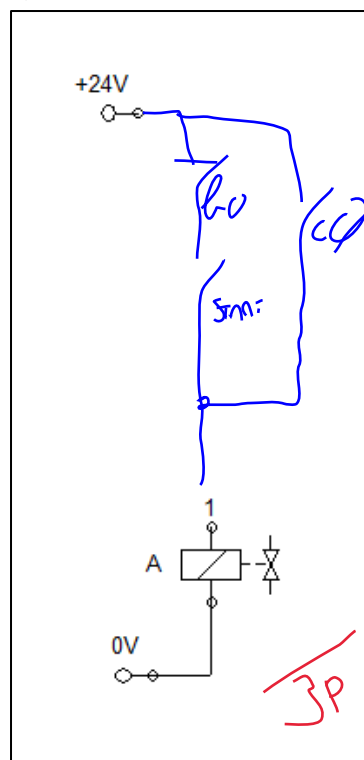
Opgave 2 - Pneumatisch schakelen

a)

$$A = \overline{b0} \cdot \text{START} + c0$$

- 1 P PER FUNCT RSU:
- OVERBODIGE HAAKJES
- FOUTE OPERANDEN 3P

b)



- 1 P PER FUNCT RSU:
- NO/NC
- EN/OF

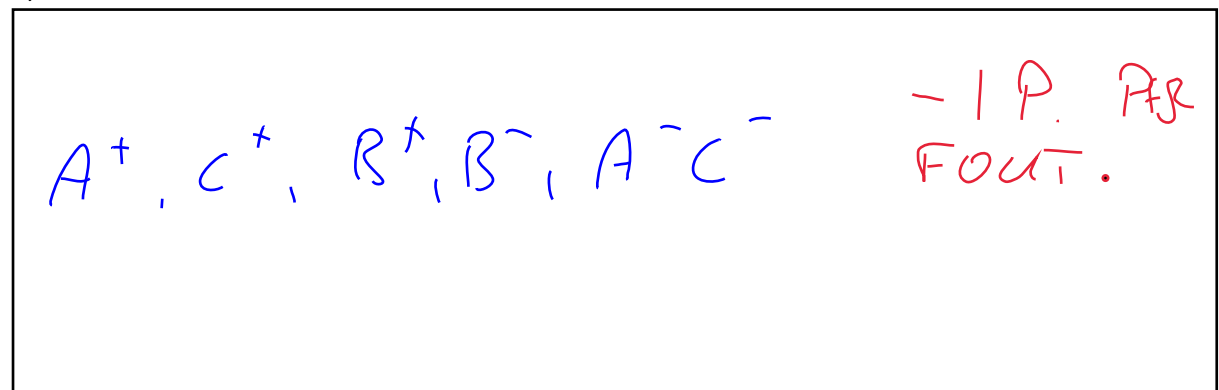
c)

b0	start	c0	A
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

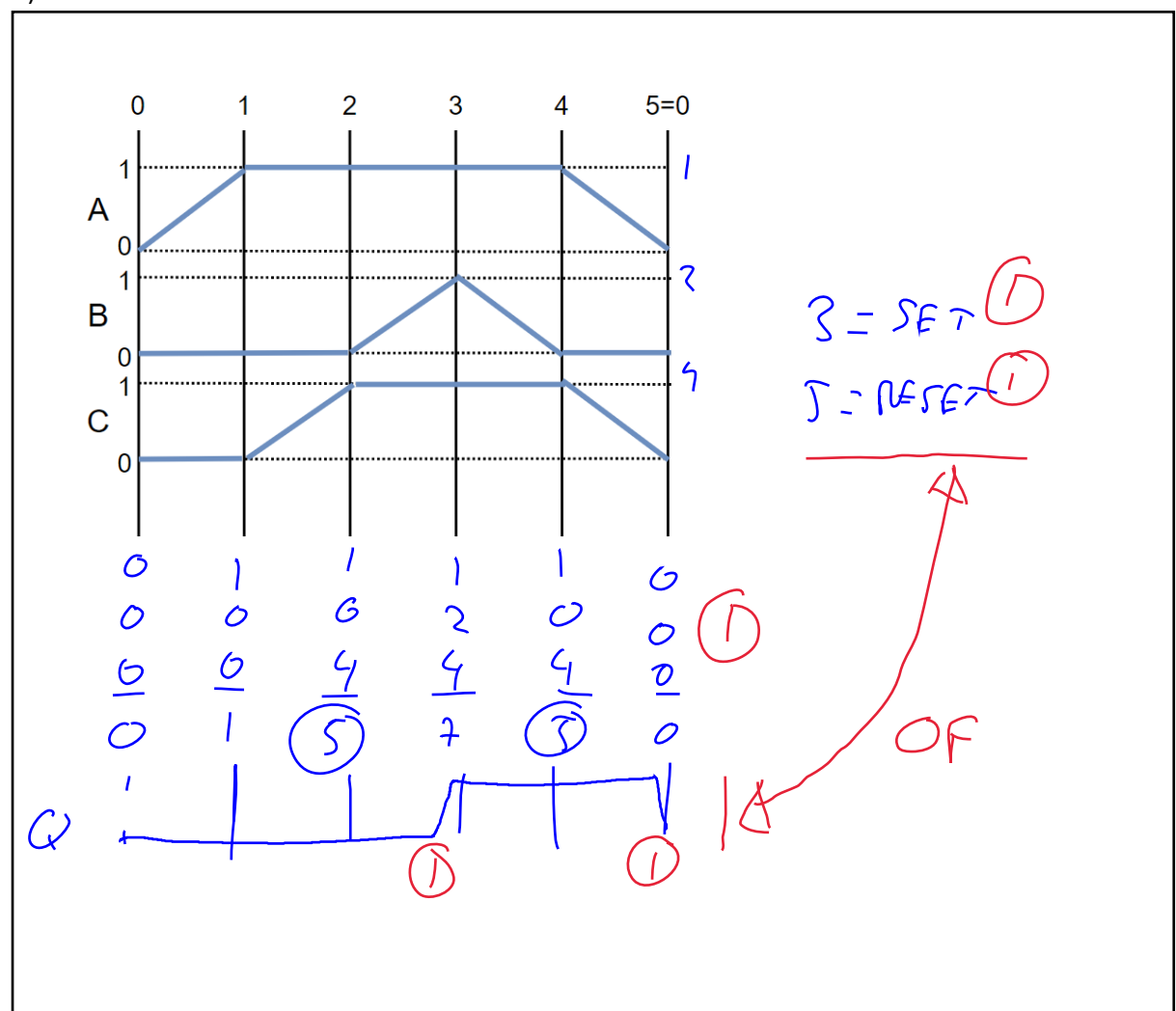
2P.

Opgave 3 – Volgordebesturing

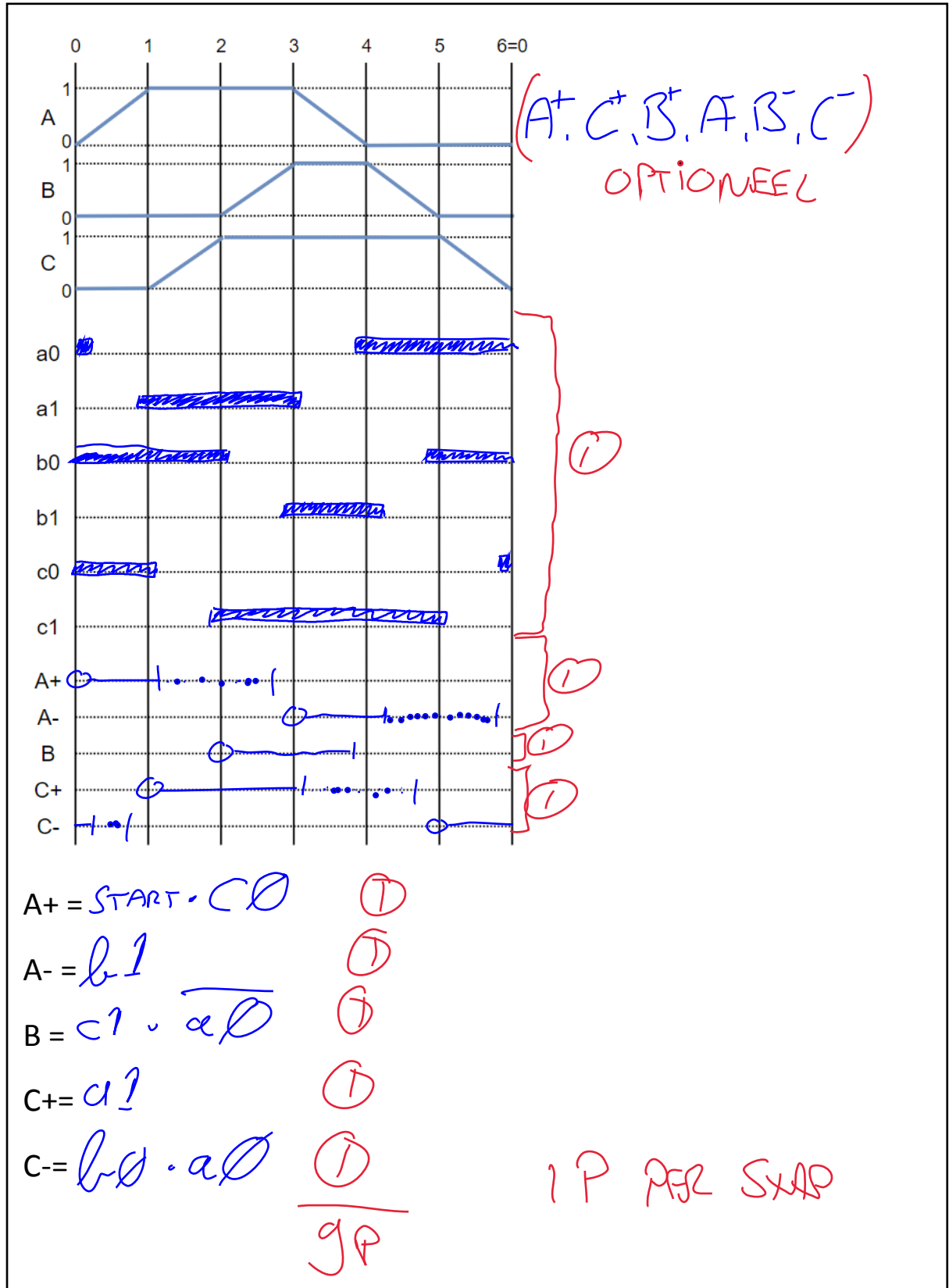
a)



b)



c)



Opgave 4 – Stromingsleer

a)

$$P = \rho \cdot g \cdot h \quad (1)$$

$$P = \underset{(1)}{1000} \cdot 10 \cdot 10 = 100 \text{ kPa} \quad (1)$$

OF
1 BAR

3P

b)

$$\underset{\mu}{Re = v \cdot L \cdot \rho} \quad (1) \quad 3500 \quad (1)$$

$$v = \frac{Re \quad (1)}{\left(\frac{L \cdot \rho}{\mu} \right)} = \frac{3500}{\left(\frac{0,03 \cdot 1000}{1,0 \cdot 10^{-2}} \right)} = 1,16 \text{ m/s} \quad (1)$$

4P

c)

$$P_w = f \cdot \frac{l}{d} \cdot \frac{1}{2} \cdot \rho \cdot v^2 \quad f = 0,022 \text{ ①}$$

$$l = 40 \text{ m} + ((60 + 60 + 200) \cdot 0,03 \text{ m}) = 49,6 \text{ m}$$

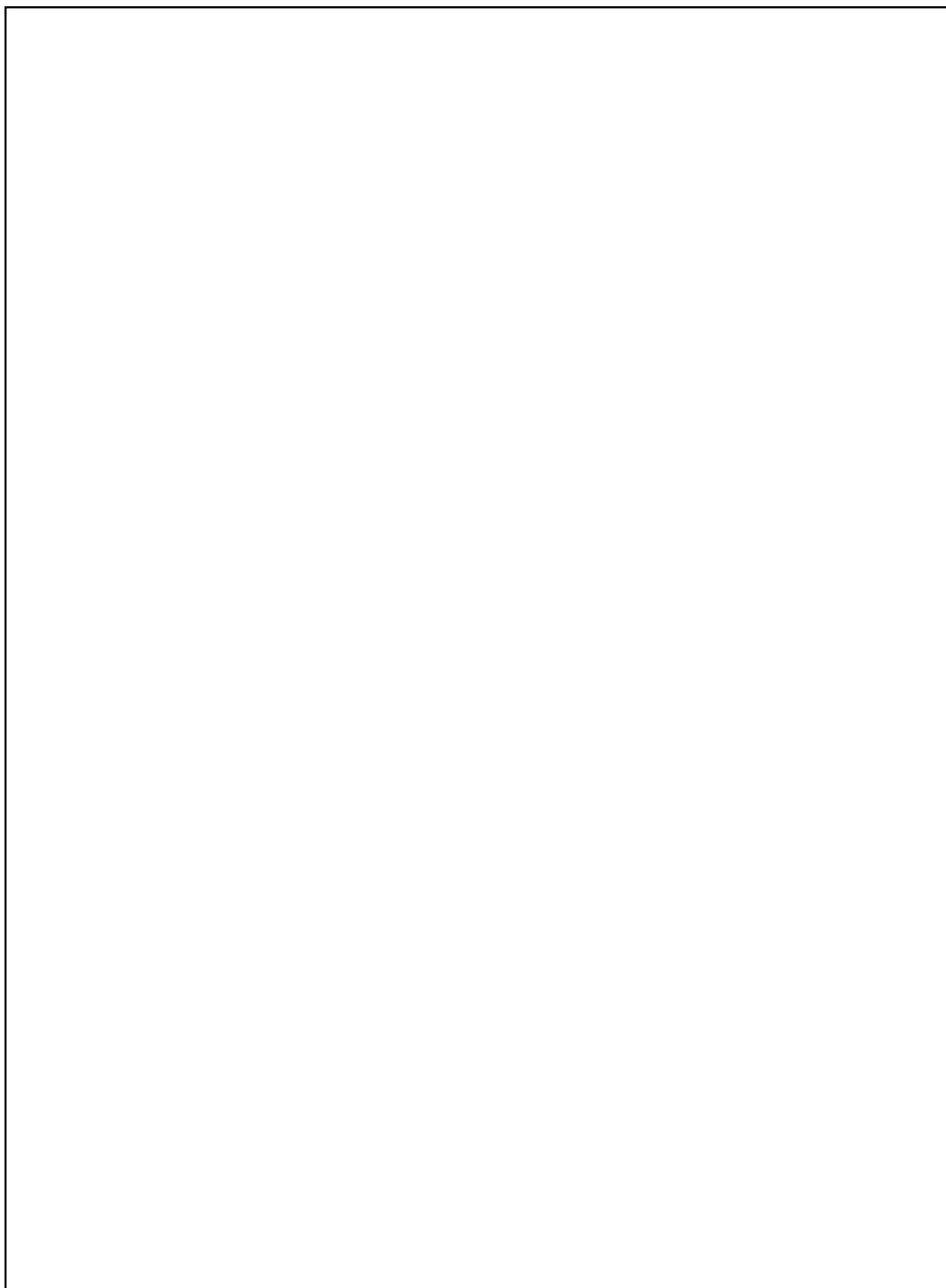
$$P_w = 0,022 \cdot \frac{49,6}{0,03} \cdot \frac{1}{2} \cdot 1000 \cdot 2^2 = 72,7 \text{ kPa}$$

①

49,6

of ①
0,72 ~~88~~

Vrije ruimte:

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